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Sheaf cohomology

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Introduction

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Part I

Category theory

Chapter 1

Categories, morphisms and duality

1.1 Categories

Definition 1.1.1 (Category)**.** A **category** C consists of:

- a collection $\mathrm{Ob}(\mathsf{C})$ of **objects**.
- for each pair of objects $X, Y \in \mathrm{Ob}(\mathsf{C})$ a collection $\mathrm{Hom}_{\mathsf{C}}(X, Y)$ of **morphisms from** X **to** Y.
- for each object $X \in \mathrm{Ob}(\mathsf{C})$ a morphisms $\mathbb{1}_X \in \mathrm{Hom}_{\mathsf{C}}(X, X)$ called the **identity** morphism.
- for each pair of morphisms $f \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ and $g \in \mathrm{Hom}_{\mathsf{C}}(Y, Z)$ a morphism $g \circ f \in \mathrm{Hom}_{\mathsf{C}}(X, Z)$ called their **composite**.

satisfying the following axioms:

- Composition is associative: for each quadruple $X, Y, Z, W \in \mathrm{Ob}(\mathsf{C})$ of objects, if $f \in \mathrm{Hom}_{\mathsf{C}}(Y, Z)$, $g \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ and $h \in \mathrm{Hom}_{\mathsf{C}}(W, X)$ then
$$(f \circ g) \circ h = f \circ (g \circ h)$$
- Composition satiesfies the unit laws: for each pair of objects $X, Y \in \mathrm{Ob}(\mathsf{C})$ if $f \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ then
$$\mathbb{1}_Y \circ f = f = f \circ \mathbb{1}_X$$

⌟

Notation. To indicate that X is an object of a category C we write $X \in \mathsf{C}$ instead of $X \in \mathrm{Ob}(\mathsf{C})$.

To indicate that $f \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ is a morphisms from X to Y we write $f : X \to Y$ or $X \xrightarrow{f} Y$.

Example 1.1.1.

- The category Set has the collection of all sets as $\mathrm{Ob}(\mathsf{Set})$ and, for any pair of sets X, Y, $\mathrm{Hom}_{\mathsf{Set}}(X, Y)$ is the set of all applications $f : X \to Y$.
- The category Grp has the collection of all groups as $\mathrm{Ob}(\mathsf{Grp})$ and, for any pair of groups G, H $\mathrm{Hom}_{\mathsf{Grp}}(G, H)$ is the set of all group homomorphisms $f : G \to H$.

- The category Top has the collection of all topological spaces as $\mathrm{Ob}(\mathsf{Top})$ and, for any pair of topological spaces X, Y $\mathrm{Hom}_{\mathsf{Top}}(X, Y)$ is the set of all continuous functions $f : X \to Y$.

◊

Definition 1.1.2 (Locally small category)**.** A category C is **locally small** if for every pair of objects $X, Y \in \mathsf{C}$ the collection $\mathrm{Hom}(X, Y)$ is a set. ⌟

Definition 1.1.3 (Small category)**.** A locally small category C is **small** if its collection of objects $\mathrm{Ob}(\mathsf{C})$ is a set. ⌟

1.2 Types of morphisms

Definition 1.2.1 (Isomorphism)**.** An **isomorphism** in a category is a morphism $f : X \to Y$ for which exists another morphism $g : Y \to X$ such that:

$$g \circ f = \mathbb{1}_X \quad \text{and} \quad f \circ g = \mathbb{1}_Y$$

The morphism g is called the **inverse** of f, often denoted f^{-1}. ⌟

Example 1.2.1.

- In Set, an isomorphism is a bijection.
- In Grp, an isomorphism is a bijective group homomorphism.
- In Top, an isomorphism is a homeomorphism.

◊

Proposition 1.2.1. *Let $f : X \to Y$, $g : Y \to Z$ be two isomorphisms in a category C. Then:*

1. *The inverse f^{-1} of f is unique.*
2. *$g \circ f$ is an isomorphism.*
3. *If $h \circ k$ is an isomorphism and either g or h is an isomorphism then so is the other.*

1.3 Duality

Definition 1.3.1 (Opposite category)**.** Let C be a category. The **opposite category** C^{op} consists of:

- The same objects as C: $\mathrm{Ob}(\mathsf{C}^{\mathrm{op}}) = \mathrm{Ob}(\mathsf{C})$.
- For each pair of objects $X, Y \in \mathsf{C}^{\mathrm{op}}$: $\mathrm{Hom}_{\mathsf{C}^{\mathrm{op}}}(X, Y) = \mathrm{Hom}_{\mathsf{C}}(Y, X)$.

⌟

Example 1.3.1. Consider a preordered set $(P, \leq)$ viewed as a small category where a unique morphism $x \to y$ exists if and only if $x \leq y$.

The opposite category P^{op} corresponds to the same set P equipped with the reversed order $\geq$. That is, a morphism $x \to y$ exists in P^{op} if and only if $y \leq x$ in P. ◊

Chapter 2

Functors and natural transformations

2.1 Functors

2.2 Natural transformations

2.3 Representable functors

Chapter 3

Limits and colimits

3.1 Diagrams, limits and colimits

3.2 Initial and terminal objects

3.3 Products and coproducts

3.4 Directed and inverse limits

Chapter 4

More on functors

4.1 Adjunctions

4.2 Functor categories

Part II

Homological algebra

Chapter 5

Abelian categories and complexes

5.1 Abelian categories

5.1.1 Additive categories

Definition 5.1.1 (Additive category)**.** A category A is **additive** if it verifies:

A.1) There exists a zero object $0 \in \mathsf{A}$.

A.2) A has finite categorical products.

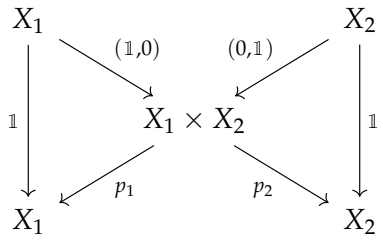
A.3) For every pair of objects $X, Y \in \mathsf{A}$ the set $\mathrm{Hom}_{\mathsf{A}}(X, Y)$ is endowed with an abelian group structure and the composition maps are bilinear with respect to the group operation.

⌟

Lemma 5.1.1. *Finite products and coproducts coincide in an additive category.*

Proof. Consider $X_1, X_2 \in \mathsf{A}$. We only need to check that $X_1 \times X_2 \cong X_1 \amalg X_2$ since the case for a finite collection of elements follows directly by induction. To do this, we will see that $X_1 \times X_2$ satisfies the universal property of the coproduct.

Consider the following diagram



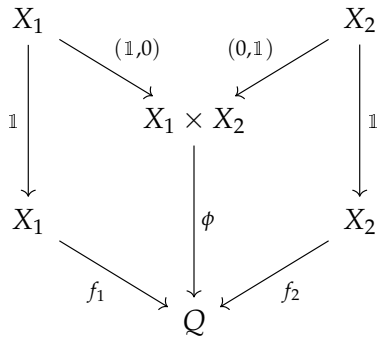
Note that for any pair of morphisms $(\alpha_1, \beta_1), (\alpha_2, \beta_2) : R \to X_1 \times X_2$ we have that

$$(\alpha_1, \beta_1) + (\alpha_2, \beta_2) = (\alpha_1 + \alpha_2, \beta_1, \beta_2)$$

Therefore if p_1, p_2 are two projections on $X_1 \times X_2$ whose sum is the identity we have that

$$(\mathbb{1}, 0) \circ p_1 + (0, \mathbb{1}) \circ p_2 = \mathbb{1}_{X_1 \times X_2}$$

Consider now two morphisms $f_i : X_i \to Q$. Then its sum $\phi = f_1 \circ p_1 + f_2 \circ p_2$ produces the following diagram



Furthermore, this is unique. Indeed, consider $\phi' : X_1 \times X_2 \to Q$ another morphism filling the diagram. Then

$$\begin{aligned}
\phi &= \phi \circ \mathbb{1}_{X_1 \times X_2} \\
&= \phi \circ ((\mathbb{1}, 0) \circ p_1 + (0, \mathbb{1}) \circ p_2) \\
&= \phi \circ (id, 0) \circ p_1 + \phi \circ (0, \mathbb{1}) \circ p_2 \\
&= f_1 \circ p_1 + f_2 \circ p_2 \\
&= \phi' \circ (\mathbb{1}, 0) + \phi' \circ (0, \mathbb{1}) \\
&= \phi' \circ ((\mathbb{1}, 0) \circ p_1 + (0, \mathbb{1}) \circ p_2) \\
&= \phi' \circ \mathbb{1}_{X_1 \times X_2} \\
&= \phi'
\end{aligned}$$

Therefore $X_1 \times X_2$ verifies the universal property of $X_1 \amalg X_2$. □

Notation. If $\{X_i\}_{i=1}^n$ is a finite set of objects in an additive category we will denote its product (and therefore by Lemma 5.1.1 its coproduct) by $\bigoplus_{i=1}^n X_i$.

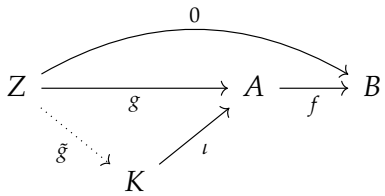
Lemma 5.1.2. *A morphism $\varphi : A \to B$ in an additive category is a monomorphism if and only if for every morphism $\xi : Z \to A$*

$$\varphi \circ \xi = 0 \implies \xi = 0$$

Dually φ is an epimorphism if and only if for every morphism $\xi : B \to Z$

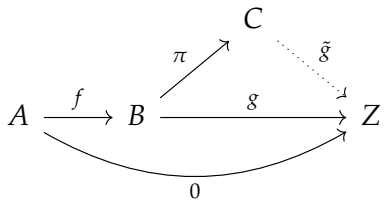
$$\xi \circ \varphi = 0 \implies \xi = 0$$

Definition 5.1.2 (Kernel)**.** Let $f : A \to B$ be a morphism in an additive category. A morphism $\iota : K \to A$ is a **kernel** of f if $f \circ \iota = 0$ and for every morphism $g : Z \to A$ such that $f \circ g = 0$ there exists a unique $\tilde{g} : Z \to K$ making the following diagram commute



⌟

Definition 5.1.3 (Cokernel)**.** Let $f : A \to B$ be a morphism in an additive category. A morphism $\pi : B \to C$ is a **cokernel** of f if $\pi \circ f = 0$ and for every morphism $g : B \to Z$ such that $\pi \circ g = 0$ there exists a unique $\tilde{g} : C \to Z$ making the following diagram commute

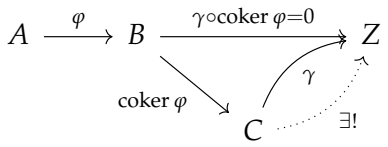


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Lemma 5.1.3. *In an additive category kernels are monomorphisms and cokernels are epimorphisms.*

Proof. We will only show that cokernels are epimorphisms. The statement about kernels follows analogously.

Let $\varphi : A \to B$ be a morphism and let $\operatorname{coker}\varphi : B \to C$ be its cokernel. Consider another morphism $\gamma : C \to Z$ such that $\gamma \circ \operatorname{coker}\varphi = 0$. Since $(\gamma \circ \operatorname{coker}\varphi) \circ \varphi = 0$ by definition of the cokernel we have



Therefore since $\gamma \circ \operatorname{coker}\varphi = 0 = 0 \circ \operatorname{coker}\varphi$ uniqueness forces $\gamma = 0$ and thus by Lemma 5.1.2 φ is an epimorphism. □

Lemma 5.1.4. *Let $\varphi : A \to B$ be a morphism in an additive category. Then φ is a monomorphism if and only if $0 \to A$ is its kernel and φ is an epimorphism if and only if $B \to 0$ is its cokernel.*

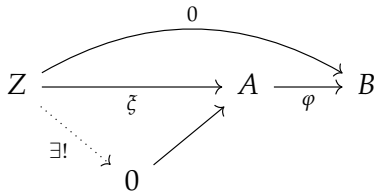
Proof. We will do the kernel one.

$\Longrightarrow$ Suppose φ is a monomorphism.

If $\xi : Z \to A$ is a morphism such that $\varphi \circ \xi = 0$ then by Lemma 5.1.2 $\xi = 0$. In particular ξ factors uniquely through $0 \to A$, i.e: $0 \to A$ is the kernel of φ.

$\Longleftarrow$ Suppose now that $0 \to A$ is the kernel of φ.

Consider a morphism $\xi : Z \to A$ such that $\varphi \circ \xi = 0$. By the kernel properties we have the following diagram



Therefore by uniqueness $\xi = 0$, making φ a monomorphism by Lemma 5.1.2.

□

Definition 5.1.4 (Additive functor)**.** Let A, B be two additive categories. A functor $F : \mathsf{A} \to \mathsf{B}$ is **additive** if it preserves the abelian group structure of all Hom-sets. ⌟

5.1.2 Abelian categories

Definition 5.1.5 (Abelian category)**.** An additive category A is **abelian** if it verifies

A.4) Kernels and cokernels exist in A.

A.5) Every monomorphism is the kernel of some morphism and every epimorphism is the cokernel of some morphism.

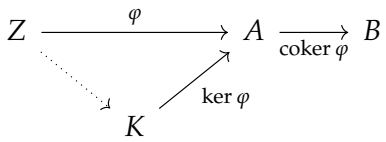
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Lemma 5.1.5. *In an abelian category* A *every kernel is the kernel of its cokernel and every cokernel is the cokernel of its kernel. In formulas*

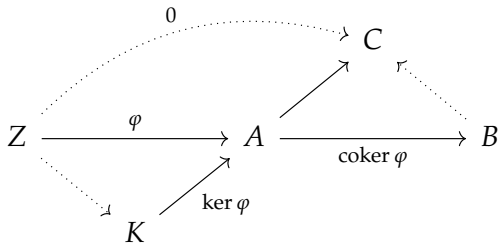
$$\ker(\operatorname{coker}\varphi) = \ker\varphi \qquad \operatorname{coker}(\ker\varphi) = \operatorname{coker}\varphi$$

Proof. Lets check cokernels this time.

Let $\operatorname{coker}\varphi : A \to B$ be the cokernel of a morphism $\varphi : Z \to A$. Since A is abelian φ has a kernel $\ker\varphi : K \to A$. Note that $Z \to A \to B$ is 0, therefore by the kernel universal property we have



Consider a morphism $A \to C$ such that $(K \to A \to C) = 0$. Now using the cokernel universal property we obtain

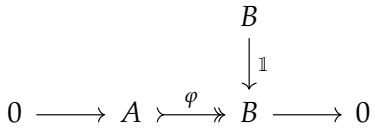


Therefore $\operatorname{coker}\varphi$ satisfies the universal property of the cokernel of $\ker\varphi : K \to A$. □

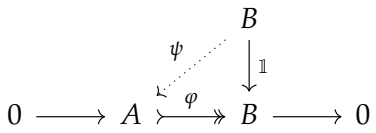
Lemma 5.1.6. *Let $\varphi : A \to B$ be a morphism in an abelian category* A*. If φ is both a monomorphism and epimorphism then φ is an isomorphism.*

Proof. By Lemma 5.1.4 the kernel of φ is $0 \to A$ and its cokernel is $B \to 0$. Furthermore by Lemma 5.1.5 φ is the cokernel of $0 \to A$ and the kernel of $B \to 0$.

Consider the following diagram

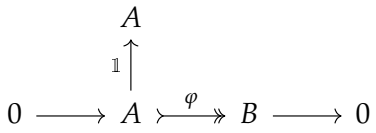


Since $(B \to B \to 0) = 0$ and φ is the kernel of $B \to 0$ we have

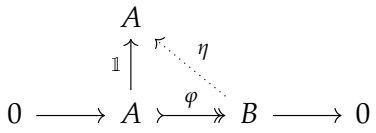


Therefore since $\varphi\psi = \mathbb{1}$ we have shown that φ has a right-inverse.

Consider now the diagram



Since $(0 \to A \to A) = 0$ and φ is the cokernel of $0 \to A$ we have



Therefore since $\eta\varphi = \mathbb{1}$ we have shown that φ has a left-inverse. □

Theorem 5.1.1 (Freyd-Mitchell theorem). *Let* A *be a small abelian category. Then there exists a fully faithfull exact functor* $\mathsf{A} \to R\text{-}\mathsf{Mod}$ *for a suitable (possibly non-commutative) ring* R*, where* R-Mod *is the category of left* R*-modules.*

5.2 Complexes and cohomology

Throughout this chapter we will be working over a fixed abelian category A.

Definition 5.2.1 (Complex). A sequence of objects and morphisms

$$\cdots \xrightarrow{d_{i-2}} C_{i-1} \xrightarrow{d_{i-1}} C_i \xrightarrow{d_i} C_{i+1} \xrightarrow{d_{i+1}} \cdots$$

in A is a **chain complex** if for every $i \in \mathbb{N} \quad d_{i-1} \circ d_i = 0$. We denote a chain complex as $(C_\bullet, d_\bullet)$ or simply as $C_\bullet$.

By reversing the order, we say that a sequence

$$\cdots \xrightarrow{d^{i+2}} C^{i+1} \xrightarrow{d^{i+1}} C^i \xrightarrow{d^i} C^{i-1} \xrightarrow{d^{i-1}} \cdots$$

is a **cochain complex** if for every $i \in \mathbb{N} \quad d^i \circ d^{i+1} = 0$. We denote a cochain complex as $(C^\bullet, d^\bullet)$ or simply as $C^\bullet$. ⌟

Remark 5.2.1. In purelly algebraic terms, chains are cochains are *equivalent*. We will adopt the cochain notation, but all proofs and definitions translate to the chain notation. ▽

Definition 5.2.2 (Exact complex). A complex

$$\cdots \xrightarrow{d^{i+2}} C^{i+1} \xrightarrow{d^{i+1}} C^i \xrightarrow{d^i} C^{i-1} \xrightarrow{d^{i-1}} \cdots$$

in A is **exact at** C^i (or simply at position i) if

$$\operatorname{coker} d^{i+1} \circ \ker d^i = 0$$

A complex is **exact** if it is exact at every position. ⌟

Definition 5.2.3 (Short exact sequence)**.** An exact complex of the form

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

is called a **short exact sequence**.

Exact complexes of the form $0 \to X \to Y \to Z$ (resp. $X \to Y \to Z \to 0$) are called left (resp. right) short exact sequences. ⌟

Remark 5.2.2. For any morphism $f : X \to Y$ in an A we get short exact sequences

$$0 \longrightarrow \ker f \longrightarrow X \longrightarrow \operatorname{im} f \longrightarrow 0$$

$$0 \longrightarrow \operatorname{im} f \longrightarrow Y \longrightarrow \operatorname{coker} f \longrightarrow 0$$

Therefore by Lemma 5.1.4 the sequence $0 \to X \xrightarrow{f} Y$ (resp. $X \xrightarrow{f} Y \to 0$) is exact if and only if f is a monomorphism (resp. epimorphism). ▽

Definition 5.2.4 (Exact functor)**.** Let A, B be two abelian categories.

An additive functor $F : \mathsf{A} \to \mathsf{B}$ is called left (resp. right) **exact** if for any short exact sequence in A

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z$$

(resp. $X \to Y \to Z \to 0$) the squence

$$0 \longrightarrow F(X) \longrightarrow F(Y) \longrightarrow F(Z)$$

(resp. $F(X) \to F(Y) \to F(Z) \to 0$) is exact.

If F is both left and right exact, then F is called **exact**. ⌟

Example 5.2.1. The functors $\operatorname{Hom}_{\mathsf{A}}(X, -)$ and $\operatorname{Hom}_{\mathsf{A}}(-, X)$ are both left-exact functors from A to Ab. ◊

The following is [Sch, Lemma 5.2.2].

Lemma 5.2.1. *Let $F : \mathsf{A} \to \mathsf{B}$ be an additive functor.*

1. *The following are equivalent*
   (a) *F is left exact.*
   (b) *F commutes with kernels: for any morphism $f : X \to Y$, $F(\ker f) \xrightarrow{\sim} \ker F(f)$.*
   (c) *F commutes with finite limits.*
   (d) *For any short exact sequence $0 \to X \to Y \to Z \to 0$ in A the sequence $0 \to F(X) \to F(Y) \to F(Z)$ is exact in B.*
2. *The following are equivalent*
   (a) *F is exact.*
   (b) *For any exact sequence $X \to Y \to Z$ in A the sequence $F(X) \to F(Y) \to F(Z)$ is exact in B.*
   (c) *F commutes with finite limits and finite colimits.*

Definition 5.2.5.

- An object $P \in \mathsf{A}$ is **projective** if all the diagrams of the form

$$\begin{array}{ccc} & & X \\ & & \downarrow \\ P & \longrightarrow & Z \end{array}$$

  where $X \twoheadrightarrow X$ is an epimorphism admit an extension

$$\begin{array}{ccc} & \nearrow & X \\ & & \downarrow \\ P & \longrightarrow & Z \end{array}$$

- An object $I \in \mathsf{A}$ is **injective** if all the diagrams of the form

$$\begin{array}{ccc} X & \longrightarrow & I \\ \downarrow & & \\ Z & & \end{array}$$

  where $X \rightarrowtail Z$ is a monomorphism admit an extension

$$\begin{array}{ccc} X & \longrightarrow & I \\ \downarrow & \nearrow & \\ Z & & \end{array}$$

⌟

Proposition 5.2.1. *An object $I \in \mathsf{A}$ is injective if and only if the functor $\mathrm{Hom}_{\mathsf{A}}(-, I)$ is exact.*

An object $P \in \mathsf{A}$ is projective if and only if the functor $\mathrm{Hom}_{\mathsf{A}}(P, -)$ is exact.

Definition 5.2.6.

- A has **enough projectives** if for every object $A \in \mathsf{A}$ there exists a projective object $P \in \mathsf{A}$ and an epimorphism $P \twoheadrightarrow A$.
- A has **enough injectives** if for every object $A \in \mathsf{A}$ there exists a injective object $I \in \mathsf{A}$ and an monomorphism $A \rightarrowtail I$.

⌟

5.3 Homotopy and Quasi-Isomorphisms

Chapter 6

Derived functors

Part III

Sheaves

Chapter 7

Presheaves and sheaves

7.1 Presheaves

We begin by giving a general definition of a presheaf.

Definition 7.1.1 (Presheaf)**.** Let X be a topological space and let $\mathsf{Op}(X)$ be its `ex:open-set-category`. Let C be a Locally small category. The Locally small category of C-valued **presheaves** is the `ex:functor-category`

$$\mathsf{PSh}_X(\mathsf{C}) := [\mathsf{Op}(X)^{\mathrm{op}}, \mathsf{C}]$$

When the topological space is clear we will write $\mathsf{PSh}(\mathsf{C})$.

The elements of this category are **??**s $\mathcal{F} : \mathsf{Op}(X)^{\mathrm{op}} \to \mathsf{C}$ and morphisms are **??**s $\phi : \mathcal{F} \to \mathcal{G}$. ⌟

Let's rewrite Definition 7.1.1 less abstractly: a C-valued Presheaf $\mathcal{F}$ is the collection of data

- For every open set $U \subseteq X$ an object $\mathcal{F}(U)$ of C called *sections of $\mathcal{F}$ over U*.
- For every inclussion of open sets $V \subseteq U$ a morphism $\mathrm{res}_{U,V} : \mathcal{F}(U) \to \mathcal{F}(V)$ in C called *restriction* such that
  - For every open set U, $\mathrm{res}_{U,U} = \mathbb{1}_U$.
  - For every inclussion of open sets $W \subseteq V \subseteq U$, $\mathrm{res}_{U,V} \circ \mathrm{res}_{V,W} = \mathrm{res}_{U,W}$.

A *morphism of Presheaves* $\phi : \mathcal{F} \to \mathcal{G}$ is a collection of morphisms $\{\phi(U) : \mathcal{F}(U) \to \mathcal{G}(U)\}_{U \in \mathsf{Op}(X)}$ in C making the following diagram commute:

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\phi(U)} & \mathcal{G}(U) \\ \downarrow{\scriptstyle \mathrm{res}_{U,V}} & & \downarrow{\scriptstyle \mathrm{res}_{U,V}} \\ \mathcal{F}(V) & \xrightarrow{\phi(V)} & \mathcal{G}(V) \end{array}$$

We are going to study now **??**s and **??**s in $\mathsf{PSh}(\mathsf{C})$.

Note that, since $\mathsf{PSh}(\mathsf{C})$ is a `def:functor-category` then if C has **??**s or **??**s of a certain shape then $\mathsf{PSh}(\mathsf{C})$ also has them and they are computed pointwise.

This means that **??**s and **??**s in $\mathsf{PSh}(\mathsf{C})$ can be computed *open set by open set*.

Proposition 7.1.1. *Let X be a topological space. Let $\mathcal{I} \to \mathsf{PSh}_X(\mathsf{C}), i \mapsto \mathcal{F}_i$ be a `def:diagram`. If C has **??**s and/or **??**s with shape $\mathcal{I}$ then $\lim_i \mathcal{F}_i$ and/or $\mathrm{colim}_i \mathcal{F}_i$ exist in $\mathsf{PSh}_X(\mathsf{C})$. Moreover*

for every open set $U \subseteq X$

$$\left(\lim_i \mathcal{F}_i\right)(U) = \lim_i \mathcal{F}_i(U) \qquad \left(\operatorname*{colim}_i \mathcal{F}_i\right)(U) = \operatorname*{colim}_i \mathcal{F}_i(U)$$

Write a section on functor categories (or quote it from somewhere) to justify this.

In particular if C is an Abelian category then Kernels and Cokernels exist and are computed *open set by open set*, making $\mathsf{PSh}(\mathsf{C})$ an Abelian category.

Remark 7.1.1. From this point onwards, we will suppose C has all **??**s and **??**s unless stated otherwise. ▽

7.1.1 Presheaves and continuous maps

Definition 7.1.2 (Direct image Presheaf)**.** Let $f : X \to Y$ be a continuous map. Let $\mathcal{F}$ be a C-valued Presheaf over X. The **direct image Presheaf** of $\mathcal{F}$ via f is the Presheaf $f_*\mathcal{F}$ defined as, for every open subset $V \subseteq Y$

$$f_*\mathcal{F}(V) = \mathcal{F}(f^{-1}(U))$$

Given $W \subseteq V \subseteq Y$ open subsets, the restriction map comes from the commutativity of the diagram

$$\begin{array}{ccc} f_*\mathcal{F}(V) & = & \mathcal{F}(f^{-1}V) \\ \vdots & & \downarrow \\ f_*\mathcal{F}(V) & = & \mathcal{F}(f^{-1}V) \end{array}$$

This construction defines a `def:functor` $f_* : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{PSh}_Y(\mathsf{C})$. ⌟

Show that the category of presheaves over a one point space *is equal* to the evaluation category.

Example 7.1.1 (Global section)**.** Let X be a topological space. Consider the one-point topological space $\{*\}$. Let $\mathcal{F}$ be a C-valued presheaf over X. The **global section** of $\mathcal{F}$ over

This defines a `def:functor` $\Gamma(X, -) : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{C}$ ◊

Example 7.1.2 (Skyscrapper Presheaf)**.** Let X be a topological space and $p \in X$. Suppose C has a final object $* \in \mathsf{C}$. Let S be an object in C. The **skyscrapper Presheaf** supported at S $\iota_{p,*}\underline{S}_{\text{pre}}$ is the Direct image Presheaf of the Constant Presheaf $\underline{S}_{\text{pre}}$ via the inclussion $\iota : \{p\} \hookrightarrow X$:

$$(\iota_{p,*}\underline{S}_{\text{pre}})(U) = \begin{cases} S & p \in U \\ * & p \notin U \end{cases}$$

◊

Using this we can construct a *dual* Presheaf to Definition 7.1.2.

Definition 7.1.3 (Inverse image Presheaf)**.** Let $f : X \to Y$ be a continuous map. Let $\mathcal{G}$ be a C-valued Presheaf over Y. The **inverse image Presheaf** of $\mathcal{G}$ via f is the Presheaf $f^{-1}\mathcal{G}$ defined as, for every open subset $U \subseteq X$

$$f^{-1}\mathcal{G}(U) = \operatorname*{colim}_{V \supseteq f(U)} \mathcal{G}(V)$$

where the def:categorical-colimit is taken over all the open neighbourhoods V of $f(U)$.

This defines a def:functor $f^{-1} : \mathsf{PSh}_Y(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$. ⌟

Remark 7.1.2. Note that the set of open neighbourhoods of $f(U)$ is a partially ordered set via reverse inclussion. Therefore if C is a concrete category we can describe $f^{-1}\mathcal{G}(U)$ as $\{(s, V) \mid f(U) \subseteq V \in \mathsf{Op}(Y), s \in \mathcal{G}(V)\}$ modulo the relation $(s, V) \sim (s', V')$ if and only if there exists $W \subseteq V \cap V'$ open with $f(U) \subseteq W$ such that $s|_W = s'|_W$. ▽

Example 7.1.3 (Constant Presheaf)**.** Let X be a topological space. Let S be an object of C. The **constant Presheaf** associated to S $\underline{S}_{\text{pre}}$ is the Inverse image Presheaf of S via the map $\{*\} \to X$. ◊

Definition 7.1.4 (Stalk)**.** Let X be a topological space and $p \in X$. Let $\mathcal{F}$ be a C-valued Presheaf over X. The **stalk** of $\mathcal{F}$ at p $\mathcal{F}_p$ is the Inverse image Presheaf of $\mathcal{F}$ via $X \to \{p\}$:

$$\mathcal{F}_p = \operatorname*{colim}_{U \ni p} \mathcal{F}(U)$$

⌟

Proposition 7.1.2. *Let $f : X \to Y$ be a continuous map. We have an* def:adjunction *between the Direct image Presheaf functor and the Inverse image Presheaf $f^{-1} \dashv f_*$.*

Proof. We define the def:categorical-unit $\eta : 1 \implies f_* f^{-1}$ as the map with components given by the canonical insertion maps to the colimit

$$\mathcal{F}(W) \to \operatorname*{colim}_{U \supseteq (f \circ f^{-1})(W)} \mathcal{F}(U) = f^{-1}\mathcal{F}(f^{-1}W) = f_* f^{-1}\mathcal{F}(W)$$

Note that this maps exist because $(f \circ f^{-1})(W) \subseteq W$.

On the other hand, note that if $f(V) \subseteq U$ then $V \subseteq (f^{-1} \circ f)(V) \subseteq f^{-1}(U)$, i.e: we have a restricion map $\mathcal{G}(f^{-1}(U)) \to \mathcal{G}(V)$. Therefore by the universal property of the colimit we have a unique map

$$f^{-1} f_* \mathcal{G}(V) = \operatorname*{colim}_{U \supseteq f(V)} \mathcal{G}(f^{-1}(U)) \to \mathcal{G}(V)$$

Using this maps we obtain a def:categorical-counit $\varepsilon : f^{-1} f_* \implies 1$.

Since by construction the compositions $\eta \circ \varepsilon$ and $\varepsilon \circ \eta$ are the corresponding identity morphisms, we have proven the adjunction. □

Two important particular cases of the adjunction Proposition 7.1.2 are

Corollary 7.1.1. *Let X be a topological space and $p \in X$. Let $\mathcal{F}$ be a C-valued Presheaves over X and let A be an object in C. Then*

$$\mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\underline{A}_{\text{pre}}, \mathcal{F}) \simeq \mathrm{Hom}_{\mathsf{C}}(A, \Gamma(X, \mathcal{F}))$$

Corollary 7.1.2. *Let X be a topological space and $p \in X$. Let $\mathcal{F}$ be a C-valued Presheaves over X and let A be an object in C. Then*

$$\mathrm{Hom}_{\mathsf{C}}(\mathcal{F}_p, A) \simeq \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\mathcal{F}, \iota_{p,*}\underline{A}_{\text{pre}})$$

7.2 Sheaves

We now turn our attention to a particular class of Presheaves, following [**stacks-project**].

Let's start by considering the case $\mathsf{C} = \mathsf{Set}$:

Definition 7.2.1 (Sheaf of sets)**.** Let X be a topological space. A Presheaf of sets $\mathcal{F}$ is a **sheaf** if for every open set U and open covering $\{U_i\}_{i\in\mathcal{I}}$ of U the diagram

$$\mathcal{F}(U) \to \prod_{i\in\mathcal{I}} \mathcal{F}(U_i) \rightrightarrows \prod_{(i,j)\in\mathcal{I}\times\mathcal{I}} \mathcal{F}(U_i \cap U_j) \tag{7.2.1}$$

is an `def:equalizer`.

We denote by $\mathsf{Sh}_X(\mathsf{Set})$ (or just Sh if the context is clear) the `def:full-subcategory` of the category of presheaves of sets whose objects are sheaves of sets.

Less abstractly this means that $\mathcal{F}$ verifies:

SH.1) **Identity**: Given two sections $s, t \in \mathscr{F}(U)$ such that for every $i \in \mathcal{I}$ $\quad s|_{U_i} = t|_{U_i}$ we have that $s = t$.

SH.2) **Gluability**: Given a collection of sections $\{s_i \mid s_i \in \mathscr{F}(U_i)\}_{i\in\mathcal{I}}$ that agree on pairwise intersections ($\forall i, j \in \mathcal{I}\ \ s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j}$) there exists a section $s \in \mathscr{F}(U)$ such that $\forall i \in \mathcal{I}\ \ s|_{U_i} = s_i$.

⌟

Definition 7.2.2 (Locally small category of Sheaves of sets)**.** The Locally small category $\mathsf{Sh}_X(\mathsf{Set})$ of Sheaves of sets is the `def:full-subcategory` of the category of Presheaves $\mathsf{PSh}_X(\mathsf{Set})$ whose objects are Sheaves of sets. ⌟

Let's try to make Definition 7.2.1 more general by considering another evaluation category C.

Firstly, note that if **??**s indexed by $\mathcal{I}$ and $\mathcal{I}\times\mathcal{I}$ for any indexing set $\mathcal{I}$ exist in C then the diagram (7.2.1) makes sense. Secondly, we have that (7.2.1) is an `def:equalizer` if and only if for every object $C \in \mathsf{C}$ the diagram in Set

$$\mathrm{Hom}_{\mathsf{C}}(C, \mathcal{F}(U)) \to \prod_{i\in\mathcal{I}} \mathrm{Hom}_{\mathsf{C}}(C, \mathcal{F}(U_i)) \rightrightarrows \prod_{(i,j)\in\mathcal{I}\times\mathcal{I}} \mathrm{Hom}_{\mathsf{C}}(C, \mathcal{F}(U_i \cap U_j))$$

is an `def:equalizer`.

Therefore we get

Definition 7.2.3 (Sheaf)**.** Let X be a topological space. Let C be an arbitrary category. A C-valued Presheaf $\mathcal{F}$ is a **sheaf** if for all objects $C \in \mathsf{C}$ the Presheaf of sets $\mathcal{F}_C$ defined as $\mathcal{F}_C(U) = \mathrm{Hom}_{\mathsf{C}}(C, \mathcal{F}(U))$ is a Sheaf of sets.

We denote by $\mathsf{Sh}_X(\mathsf{C})$ the `def:full-subcategory` of the category of C-valued presheaves of sets whose objects are C-valued sheaves. ⌟

Notation. From now on, to distinguish a Presheaf from a Sheaf we will denote the former by $\mathcal{F}, \mathcal{G}, \mathcal{H}, \ldots$ and the latter by $\mathscr{F}, \mathscr{G}, \mathscr{H}, \ldots$

By Definition 7.2.3 we have an inclussion functor $\iota : \mathsf{Sh}(\mathsf{C}) \hookrightarrow \mathsf{PSh}(\mathsf{C})$.

For *sufficiently nice* Locally small categorys, this `def:functor` admits a left-adjoint $\mathsf{PSh}(\mathsf{C}) \to \mathsf{Sh}(\mathsf{C})$. Let's study first the case $\mathsf{C} = \mathsf{Set}$.

How general do we want to go?

Theorem 7.2.1 (Sheafification). *There exists a* ***sheafification*** `def: functor` $(-)^{\#} : \mathsf{PSh}(\mathsf{Set}) \to \mathsf{Sh}(\mathsf{Set})$ *that is left-adjoint to the inclussion* `def: functor` $\iota : \mathsf{Sh}(\mathsf{Set}) \to \mathsf{PSh}(\mathsf{Set})$.

Proof. Let $\mathcal{F}$ be a Presheaf of sets over X. For every open subset $U \subseteq X$ we define

$$\mathcal{F}^{\#}(U) = \left\{ f_p \in \prod_{p \in U} \mathcal{F}_p \text{ verifying } (*) \right\}$$

where $(*)$ is the property *"for all $p \in U$ there exists an open neighbourhood $V \subseteq U$ of p and a section $s \in \mathcal{F}(V)$ such that, for every $q \in V$, $s_q = f_q$"*.

Taking $\prod_{p \in U} \mathcal{F}_p \to \prod_{q \in V} \mathcal{F}_q$ as the restriction maps we see that $\mathcal{F}^{\#}$ is a Presheaf. Let's see now that $\mathcal{F}^{\#}$ is a Sheaf. To do this, consider an open subset $U \subseteq X$ with open cover $\{U_i\}_{i \in \mathcal{I}}$.

SH.1 Let $s, t \in \mathcal{F}^{\#}(U)$ be two sections such that, for all $i \in \mathcal{I}$ $s|_{U_i} = t|_{U_i}$.

Then since for every $p \in U$ there exists $i \in \mathcal{I}$ such that $p \in U_i$ we have that $s_p = (s|_{U_i})_p = (t|_{u_i})_p = t_p$. Therefore $s = t$.

SH.2 Let $\{s_i \in \mathcal{F}^{\#}(U_i)\}_{i \in \mathcal{I}}$ be a collection of sections such that $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all $i, j \in \mathcal{I}$.

Consider $s \in \prod_{p \in U} \mathcal{F}_p$ defined as $s_p = (s_i)_p$ if $p \in U_i$. Note that this is well-defined by property $(*)$. To see that s verifies $(*)$ note that for every $p \in U_i$ there exists an open neighbourhood $V \subseteq U$ of p and $t \in \mathcal{F}(V)$ such that, for all $q \in V$, $t_q = (s_i)|_q = s_q$.

Let $V \subseteq U \subseteq X$ be two open sets. Consider the commutative diagram

$$\begin{array}{ccccc} \mathcal{F}(U) & \longrightarrow & \mathcal{F}^{\#}(U) & \longrightarrow & \prod_{p \in U} \mathcal{F}_p \\ \downarrow & & \downarrow & & \downarrow \\ \mathcal{F}(V) & \longrightarrow & \mathcal{F}^{\#}(V) & \longrightarrow & \prod_{q \in V} \mathcal{F}_q \end{array}$$

where the vertical arrows are the corresponding restriction maps. By chasing this diagram we get a map $\varphi : \mathcal{F} \to \mathcal{F}^{\#}$, $\varphi_U(s) = (s_p)_{p \in U}$.

It only remains to show that any morphism $\psi : \mathcal{F} \to \mathcal{G}$, where $\mathcal{G}$ is a Sheaf of sets, factors uniquely as $\mathcal{F} \xrightarrow{\varphi} \mathcal{F}^{\#} \to \mathcal{G}$, i.e: we have to construct a morphism $\theta : \mathcal{F}^{\#} \to \mathcal{G}$ such that $\psi = \theta \circ \varphi$.

Let $U \subseteq X$ be an open subset and $s \in \mathcal{F}^{\#}(U)$. For each $p \in U$ consider an open neighbourhood $V^{(p)} \subseteq U$ of p and $t^{(p)} \in \mathcal{F}(V^{(p)})$ such that $t^{(p)}_q = s_q$. Consider now $\psi_{V^{(p)}}(t^{(p)}) \in \mathcal{G}(V^{(p)})$. Note that $\{\psi_{V^{(p)}}(t^{(p)})\}_{p \in U}$ is a compatible family of section in $\mathcal{G}$, therefore by the Sheaf axioms there exists a unique section $t \in \mathcal{G}(U)$ such that $t|_{V^{(p)}} = \psi_{V^{(p)}}(s^{(p)})$. Therefore we can define $\theta_U(s) = t$.

By construction $\psi = \theta \circ \varphi$ and θ commutes with the restriction maps, making it a morphism of sheaves.

To check uniqueness, let $\theta' : \mathcal{F}^{\#} \to \mathcal{G}$ be another morphism such that $\psi = \theta' \circ \varphi$. Consider $s \in \mathcal{F}^{\#}(U)$. For each $p \in U$ choose $V^{(p)} \subseteq U$ and $s^{(p)} \in \mathcal{F}(V^{(p)})$ such that $s|_{V^{(p)}} = \varphi_{V^{(p)}}(s^{(p)})$. Then

$$\theta'(s)|_{V^{(p)}} = \theta'(s|_{V^{(p)}}) = \theta'(\varphi_{V^{(p)}}(s^{(p)})) = \psi_{V^{(p)}}(s^{(p)})$$

On the other hand $\theta(s)|_{V^{(p)}} = \psi_{V^{(p)}}(s^{(p)})$ by definition. Therefore since $\mathscr{G}$ is a Sheaf by SH.1 we have $\theta'(f) = \theta(f)$, i.e: $\theta' = \theta$. □

We can extend the construction outlined in Theorem 7.2.1 to **??**s.

Lemma 7.2.1. *Let (C, F) be an* `def:algebraic-category`*. Let $\mathcal{F}$ be a* C*-valued Presheaf. Then there exists a* C*-valued Sheaf $\mathcal{F}^{\#}$ and a morphism of presheaves $\mathcal{F} \to \mathcal{F}^{\#}$ satisfying*

1. *The map $\mathcal{F} \to \mathcal{F}^{\#}$ identifies the underlying Sheaf of sets of $\mathcal{F}^{\#}$ with the sheafification of the underlying Presheaf of sets of $\mathcal{F}$.*
2. *For any morphism of presheaves $\mathcal{F} \to \mathscr{G}$ where $\mathscr{G}$ is a* C*-valued Sheaf there exists a unique factorization $\mathcal{F} \to \mathcal{F}^{\#} \to \mathscr{G}$.*

Remark 7.2.1. We will assume that C is an `def:algebraic-category` in what follows. ▽

Therefore we can assign a Sheaf to any Presheaf using this construction.

Definition 7.2.4 (Sheafification)**.** Let $\mathcal{F}$ be a C-valued Presheaf. We call the Sheaf $\mathcal{F}^{\#}$ obtained by Theorem 7.2.1 the **sheafification** of $\mathcal{F}$. ⌟

Using this adjunction we can *adapt* Definition 7.1.2 and Definition 7.1.3 to the case of Sheaves

Definition 7.2.5 (Direct image Sheaf)**.** Let $f : X \to Y$ be a continuous map. Let $\mathscr{F}$ be a C-valued Sheaf over X. The **direct image Sheaf** of $\mathscr{F}$ via f is the Sheaf $f_*\mathscr{F}$ defined as, for every open subset $V \subseteq Y$

$$f_*\mathscr{F}(V) = \mathscr{F}(f^{-1}(U))$$

This construction defines a `def:functor` $f_* : \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{Sh}_Y(\mathsf{C})$. ⌟

This means that Example 7.1.1 and Example 7.1.2 translate directly to Sheaves.

Definition 7.2.6 (Global section `def:functor`)**.** We denote by $\Gamma(X, -) : \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{C}$ the **global section functor** described in Example 7.1.1, but now with initial category $\mathsf{Sh}_X(\mathsf{C})$. ⌟

Definition 7.2.7 (Skyscrapper Sheaf)**.** Let X be a topological space and $p \in X$. Let S be an object of C. The **skyscrapper Sheaf** supported at p $\iota_{*,p}\underline{S}$ is the Direct image Sheaf of the Constant Sheaf $\underline{S}$ via the map $\{p\} \to X$. ⌟

Rewrite this definitions

Explain why we need to sheafificate

Definition 7.2.8 (Inverse image Sheaf)**.** Let $f : X \to Y$ be a continuous map. Let $\mathscr{G}$ be a C-valued Sheaf over Y. The **inverse image Sheaf** of $\mathscr{G}$ via f is the Sheaf $f^{-1}\mathscr{G}$ defined as, for every open subset $U \subseteq X$

$$f^{-1}\mathscr{G} = (f^{-1}\mathscr{G})^{\#}$$

where the RHS of the equation is the Sheafification of the Inverse image Presheaf of $\mathscr{G}$ via f.

This defines a `def:functor` $f^{-1} : \mathsf{PSh}_Y(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$. ⌟

Definition 7.2.9 (Constant Sheaf)**.** Let S be an object of C. The **constant Sheaf** $\underline{S}$ associated to S is the Sheafification of the Constant Presheaf $\underline{S}_{\text{pre}}$, i.e: $\underline{S} = (\underline{S}_{\text{pre}})^{\#}$. ⌟

Definition 7.2.10 (Stalk)**.** Let X be a topological space and $p \in X$. Let $\mathscr{F}$ be a C-valued Sheaf over X. The **stalk** of $\mathscr{F}$ at p $\mathscr{F}_p$ is the Inverse image Sheaf of $\mathscr{F}$ via $X \to \{p\}$:

$$\mathscr{F}_p = \operatorname*{colim}_{U \ni p} \mathscr{F}(U)$$

⌟

Using the adjointness we have the analogous of Proposition 7.1.2

Proposition 7.2.1. *Let $f : X \to Y$ be a continuous map. We have an* `def:adjunction` *between the Direct image Sheaf functor and the Inverse image Sheaf $f^{-1} \dashv f_*$.*

Proof.

$$\begin{aligned}\operatorname{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(f^{-1}\mathscr{G}, \mathscr{F}) &= \operatorname{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(f^{-1}\mathscr{G}, \mathscr{F}) \\ &= \operatorname{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\mathscr{G}, f_*\mathscr{G}) \\ &= \operatorname{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(\mathscr{G}, f_*\mathscr{G})\end{aligned}$$

□

Therefore adjunctions Corollary 7.1.1 and Corollary 7.2.2 also happen when considering Sheaves.

Corollary 7.2.1. *Let X be a topological space and $p \in X$. Let $\mathscr{F}$ be a C-valued Sheaves over X and let A be an object in C. Then*

$$\operatorname{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(\underline{A}, \mathscr{F}) \simeq \operatorname{Hom}_{\mathsf{C}}(A, \Gamma(X, \mathscr{F}))$$

Corollary 7.2.2. *Let X be a topological space and $p \in X$. Let $\mathscr{F}$ be a C-valued Sheaves over X and let A be an object in C. Then*

$$\operatorname{Hom}_{\mathsf{C}}(\mathscr{F}_p, A) \simeq \operatorname{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(\mathscr{F}, \iota_{p,*}\underline{A})$$

For cohomology we will be interested in the case $\mathsf{C} = \mathsf{Ab}$.

Lemma 7.2.2. *Let $a : \mathsf{A} \to \mathsf{B}$ and $b : \mathsf{B} \to \mathsf{A}$ be two* **??***s. Suppose that*

1. A *and* B *are additive categories, a, b are additive functors and $b \dashv a$.*
2. B *is an Abelian category and b is left-exact.*
3. $ba \cong \mathbb{1}_{\mathsf{A}}$.

Then A *is abelian.*

Proof. Since B is an Abelian category all **??**s and **??**s exist en B.

Since b is left-adjoint it is right-exact and therefore, exact.

Let $\varphi : B_1 \to B_2$ be a morphism in B. If $K = \ker(\varphi)$ then K is the `def:equalizer` of the diagram

$$K \longrightarrow B_1 \underset{0}{\overset{\varphi}{\rightrightarrows}} B_2$$

therefore applying b we get

$$bK \longrightarrow bB_1 \underset{0}{\overset{b\varphi}{\rightrightarrows}} bB_2$$

i.e: bK is the Kernel of $b\varphi$.

Dually we get that $b\operatorname{coker}(B_1 \to B_2)$ is the Cokernel of $b\varphi$.

Therefore we have that every morphism $b\varphi$ in A admits a Kernel and a Cokernel. Moreover since $ba \cong \mathbb{1}_{\mathsf{A}}$ every morphism in A is of the form $b\varphi$, therefore all Kernels and Cokernels exist in A. Indeed for every $\psi : A_1 \to A_2$ we have that

$$\ker \psi = b\ker(a\psi) \qquad \operatorname{coker} \psi = b\operatorname{coker}(a\psi) \tag{7.2.2}$$

It only remains to show that $\operatorname{coim} \psi = \operatorname{im} \psi$. Since A has all **??**s and **??**s A has all finite **??**s and **??**s, making a left-exact. This means that a transforms Kernels into Kernels. Therefore

$$\begin{aligned}
\operatorname{coim} \psi &= \operatorname{coker}(\ker \psi \to A_1) && \text{(definition)} \\
&= b\operatorname{coker}(a(\ker \psi \to A_1)) && \text{(7.2.2)} \\
&= b\operatorname{coker}(\ker(a\psi) \to aA_1) && (a \text{ preserves kernels}) \\
&= b\operatorname{coim}(a\psi) && \text{(definition)} \\
&= b\operatorname{im}(a\psi) && (\mathsf{B} \text{ is abelian}) \\
&= b\ker(aA_2 \to \operatorname{coker}(a\psi)) && \text{(definition)} \\
&= \ker(baA_2 \to b\operatorname{coker}(a\psi)) && (b \text{ preserves kernels}) \\
&= \ker(A_2 \to b\operatorname{coker}(a\psi)) && ba \cong \mathbb{1}_{\mathsf{A}} \\
&= \ker(A_2 \to \operatorname{coker} \psi) && \text{(7.2.2)} \\
&= \operatorname{im} \psi && \text{(definition)}
\end{aligned}$$

□

Proposition 7.2.2. *Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism in $\mathsf{Sh}(\mathsf{Ab})$.*

1. *The Locally small category $\mathsf{Sh}_X(\mathsf{Ab})$ is an Abelian category.*
2. *The Kernel of φ is the same as the Kernel of φ in $\mathsf{PSh}(\mathsf{Ab})$.*
3. *φ is injective if and only if φ is injective as a map or presheaves if and only if φ is injective as a map of sheaves.*
4. *The Cokernel of φ is the Sheafification of the Cokernel of φ in $\mathsf{PSh}(\mathsf{Ab})$.*
5. *φ is surjective if and only if φ is surjective as a map of sheaves.*
6. *A complex $\mathscr{F} \to \mathscr{G} \to \mathscr{H}$ is exact at $\mathscr{G}$ if and only if for all open sets U and sections $s \in \mathscr{G}(U)$ mapping to 0 in $\mathscr{H}(U)$ there exists an open covering $\{U_i\}_{i\in\mathcal{I}}$ such that each $s|_{U_i}$ is the image of $\mathscr{F}(U_i) \to \mathscr{G}(U_i)$.*

Proof. Note that 2. $\implies$ 3., 4. $\implies$ 5 and 2. $\wedge$ 5. $\implies$ 6.

Therefore we only need to prove 1., 2. and 3., but this comes directly from Lemma 7.2.2 with $\mathsf{A} = \mathsf{Sh}_X(\mathsf{Ab})$, $\mathsf{B} = \mathsf{PSh}_X(\mathsf{Ab})$, $a : \mathsf{Sh}_X(\mathsf{Ab}) \hookrightarrow \mathsf{PSh}_X(\mathsf{Ab})$ the inclussion `def:functor` and $b : \mathsf{PSh}_X(\mathsf{Ab}) \to \mathsf{Sh}_X(\mathsf{Ab})$ the Sheafification `def:functor`. □

Chapter 8

Ringed spaces and $\mathscr{O}_X$-modules.

8.1 Ringed spaces

Definition 8.1.1 (Ringed space)**.** A **ringed space** is a tuple $(X, \mathscr{O}_X)$ where X is a topological space and $\mathscr{O}_X$ is a Sheaf of rings over X.

We call $\mathscr{O}_X$ the *structure sheaf* of the ringed space $(X, \mathscr{O}_X)$. Sections of $\mathscr{O}_X$ over an open subset $U \subseteq X$ are called *functions on* U and functions on X are called (global) functions. ⌟

Example 8.1.1. Let and X be a topological space. Then the $\mathbb{R}$-Algebra of continuous functions $\mathcal{C}^0_X(U) = \{f : U \to \mathbb{R} \mid f \text{ continuous}\}$ with the usual restriction maps forms a Sheaf of rings over X. Therefore $(X, \mathcal{C}^0_X)$ is a Ringed space. ◊

Example 8.1.2. If for every point $p \in X$ the Stalks $\mathscr{O}_{X,p}$ of the structure sheaf are Local rings then we call $(X, \mathscr{O}_X)$ a **locally ringed space**. ◊

Definition 8.1.2 (Morphism of Ringed spaces)**.** Let $(X, \mathscr{O}_X)$ and $(Y, \mathscr{O}_Y)$ be two Ringed spaces. A **morphism of ringed spaces** $(X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ is a pair $(f, f^\sharp)$ where $f : X \to Y$ is a continuous function (i.e: a morphism in Top) and $f^\sharp : \mathscr{O}_Y \to f_*\mathscr{O}_X$ is a morphism of sheaves of rings (i.e: a morphism in $\mathsf{Sh}_{\mathsf{Ring}}(X)$), where $f_*\mathscr{O}_X$ is the Direct image Sheaf. ⌟

Maybe it is better to define $f^\sharp$ as an f-map. **[stacks-project]**

Example 8.1.3. Let $f : X \to Y$ be a continuous function. Consider the Ringed spaces of continuous functions $(X, \mathcal{C}^0_X)$, $(Y, \mathcal{C}^0_Y)$.

Then the Morphism of Ringed spaces induced by f can be defined for every $V \subseteq Y$ open as $f^\sharp(V) : \mathcal{C}^0_Y(V) \to \mathcal{C}^0_X(f^{-1}(V))$, $f^\sharp(V)(h) = h \circ f|_{f^{-1}V}$. ◊

8.2 $\mathscr{O}_X$ modules

Definition 8.2.1 ($\mathscr{O}_X$-module)**.** Let $(X, \mathscr{O}_X)$ be a Ringed space. A **Sheaf of $\mathscr{O}_X$ modules** (or simply an $\mathscr{O}_X$-module) is a Sheaf $\mathscr{F}$ such that for every open set $U \subseteq X$ $\mathscr{F}(U)$ is an $\mathscr{O}_X(U)$-Module compatible with the restriction maps, i.e: for every $f \in \mathscr{O}_X(U)$ and $s \in \mathscr{F}(U)$: $(f \cdot s)|_V = f|_V \cdot s|_V$. ⌟

Example 8.2.1. ... ◊

Definition 8.2.2 (Morphism of $\mathscr{O}_X$-modules)**.** Let $\mathscr{F}, \mathscr{G}$ be two $\mathscr{O}_X$-modules. A morphism $\phi : \mathscr{F} \to \mathscr{G}$ is a sheaf morphism that commutes with the multiplication of the structure sheaf $\mathscr{O}_X$. ⌟

Example 8.2.2. ... ◊

Therefore for every Ringed space $(X, \mathscr{O}_X)$ the collection of $\mathscr{O}_X$-modules forms a Locally small category $\mathscr{O}_X$-Mod.

8.2.1 $\mathscr{O}_X$-Mod and sheaves of abelian groups.

Just as abelian groups are $\mathbb{Z}$-modules, Sheaves of abelian groups can be seen as $\underline{\mathbb{Z}}_X$-modules, where $\underline{\mathbb{Z}}_X$ is the `ex:constant-sheaf` with coefficients in the integers.

To show this, for every Sheaf of abelian groups $\mathscr{F}$ over X and every open set $U \subseteq X$ we have to define an action $\cdot : \underline{\mathbb{Z}}_X(U) \times \mathscr{F}(U) \to \mathscr{F}(U)$ compatible with the restriction maps.

Let $f \in \underline{\mathbb{Z}}_X(U)$ and $s \in \mathscr{F}(U)$. Since f is locally constant there exists an open cover $\{U_i\}_{i\in\mathcal{I}}$ of U such that for every $i \in \mathcal{I} \quad f|_{U_i} = n_i \in \mathbb{Z}$.

For every U_i consider $t_i = n_i \cdot s|_{U_i}$, where $n_i \cdot s|_{U_i}$ is shorthand for $s|_{U_i}$ added to itself n_i times. Note that this is well-defined because $\mathscr{F}(U_i)$ is an abelian group.

Let $U_i, U_j \in \{U_i\}_{i\in\mathcal{I}}$ such that $U_i \cap U_j = \varnothing$. For every $x \in U_i \cap U_j$ there exists an open neighbourhood $V \subseteq U_i \cap U_j$ of x were f is constant.

Therefore $n_i = n_j$ and $t_i|_{U_i\cap U_j} = t_j|_{U_i\cap U_j}$ and, applying axiom SH.2, we can define $f \cdot s = t$.

Checking the axioms in small enough open sets such that f is constant verifies that this action makes $\mathscr{F}$ an $\underline{\mathbb{Z}}_X$-module.

Do we have to actually write the check?

8.3 $\mathscr{O}_X$-Mod is abelian.

Let's check first that $\mathscr{O}_X$-Mod is a Additive category.

Lemma 8.3.1. *The category $\mathscr{O}_X$-Mod admits a zero object.*

Proof. Consider the $\mathscr{O}_X$ module that sends every open set to $\{0\}$. □

Lemma 8.3.2. *The category $\mathscr{O}_X$-Mod has all binary biproducts. We define their direct sum a $\mathscr{F} \oplus \mathscr{G} = \mathscr{F} \times \mathscr{G}$.*

Proof. Let $\mathscr{F}, \mathscr{G}$ be two $\mathscr{O}_X$-modules. □

Theorem 8.3.1. *Let $\varphi : \mathscr{F} \to \mathscr{G}$ a Morphism of $\mathscr{O}_X$-modules. Then the kernel of φ is the subsheaf of $\mathscr{F}$ with sections*

$$(\ker\varphi)(U) = \{s \in \mathscr{F}(U) \mid \varphi(s) = 0\}$$

Furthermore for every $x \in X \quad (\ker\varphi)_x = \ker\varphi_x$.

Proof. If $V \subseteq U$

$$\begin{array}{ccccccc} 0 & \longrightarrow & (\ker\varphi)(U) & \longrightarrow & \mathscr{F}(U) & \xrightarrow{\varphi(U)} & \mathscr{G}(U) \\ & & \vdots & & \downarrow{\scriptstyle \mathrm{res}_{U,V}} & & \downarrow{\scriptstyle \mathrm{res}_{U,V}} \\ 0 & \longrightarrow & (\ker\varphi)(V) & \longrightarrow & \mathscr{F}(V) & \xrightarrow{\varphi(V)} & \mathscr{G}(V) \end{array}$$

□

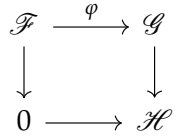
Theorem 8.3.2. *Let $\varphi : \mathscr{F} \to \mathscr{G}$ a Morphism of $\mathscr{O}_X$-modules. Then the cokernel of φ is the Sheafification of the quotient Presheaf*

$$(\operatorname{coker}_{\text{pre}} \varphi)(U) = {}^{\mathscr{G}(U)}\!/_{\varphi(\mathscr{F}(U))}$$

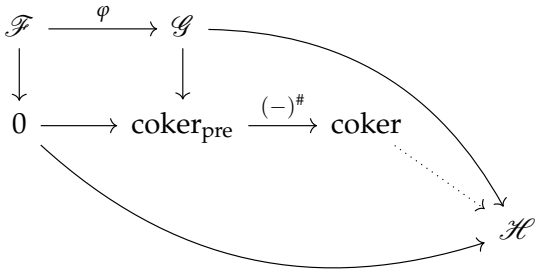
Furthermore for every $x \in X$ $\quad (\operatorname{coker} \varphi)_x = \operatorname{coker} \varphi_x$.

Prove that this is the cokernel in PSh.

Proof. Given a commutative diagram



We construct



□

Therefore we have proven that $\mathscr{O}_X$-Mod is an Abelian category and in particular by $\mathscr{O}_X$-Mod and sheaves of abelian groups $\mathsf{Sh}_{\mathsf{Ab}}(X)$ is an Abelian category.

8.4 Sequences in $\mathscr{O}_X$-Mod

Theorem 8.4.1. *The sequence of $\mathscr{O}_X$-modules $\mathscr{F} \to \mathscr{G} \to \mathscr{H}$ is exact at $\mathscr{G}$ if and only if for every $x \in X$ the sequence $\mathscr{F}_x \to \mathscr{G}_x \to \mathscr{H}_x$ is exact at $\mathscr{G}_x$.*

Probably not a *theorem* but a direct consequence of the commutatitivity of stalks and (co)kernels

Maybe talk about resolutions / types of resolutions here?

Appendix A

Algebra

Definition A.0.1 (Algebra)**.** ⌟

Definition A.0.2 (Local ring)**.** ⌟

Definition A.0.3 (Module)**.** ⌟

Bibliography

[Alu09] Paolo Aluffi. *Algebra: Chapter 0*. Graduate Studies in Mathematics v. 104. Providence, R.I: American Mathematical Society, 2009. 713 pp. ISBN: 978-0-8218-4781-7. URL: https://agorism.dev/book/math/alg/algebra_chapter-0_paolo-aluffi.pdf.

[AM18] Michael Francis Atiyah and Ian G. Macdonald. *Introduction to Commutative Algebra*. Addison-Wesley Series in Mathematics. Boca Raton London New York: CRC Press, Taylor & Francis Group, 2018. 128 pp. ISBN: 978-0-201-00361-1 978-0-201-40751-8.

[GM03] Sergej I. Gelfand and Jurij I. Manin. *Methods of Homological Algebra*. Second edition. Springer Monographs in Mathematics. Berlin, [Germany]: Springer, 2003. 1 p. ISBN: 978-3-662-12492-5. URL: https://sci-hub.ru/10.1007/978-3-662-12492-5.

[Gre+08] Gert-Martin Greuel et al. *A Singular Introduction to Commutative Algebra*. 2. , extended ed. Berlin Heidelberg: Springer, 2008. 689 pp. ISBN: 978-3-540-73541-0 978-3-642-44254-4.

[Gro] Alexandre Grothendieck. "Some Aspects of Homological Algebra". In: ().

[Har12] Günter Harder. *Lectures on Algebraic Geometry I*. 2nd revised Edition. SpringerLink Bücher. Wiesbaden: Vieweg+Teubner Verlag, 2012. 299 pp. ISBN: 978-3-8348-1844-7 978-3-8348-8330-8. DOI: 10.1007/978-3-8348-8330-8.

[Har11] Günter Harder. *Lectures on Algebraic Geometry II*. Wiesbaden: Vieweg+Teubner, 2011. ISBN: 978-3-8348-0432-7 978-3-8348-8159-5. DOI: 10.1007/978-3-8348-8159-5. URL: http://link.springer.com/10.1007/978-3-8348-8159-5 (visited on 10/16/2025).

[Hle] Theo Bu Hler. "AN INTRODUCTION TO THE DERIVED CATEGORY". In: ().

[KS06] Masaki Kashiwara and Pierre Schapira. *Categories and Sheaves*. Red. by M. Berger et al. Vol. 332. Grundlehren Der Mathematischen Wissenschaften. Berlin, Heidelberg: Springer Berlin Heidelberg, 2006. ISBN: 978-3-540-27949-5 978-3-540-27950-1. DOI: 10.1007/3-540-27950-4. URL: http://link.springer.com/10.1007/3-540-27950-4 (visited on 09/28/2025).

[KS90] Masaki Kashiwara and Pierre Schapira. *Sheaves on Manifolds*. Red. by M. Artin et al. Vol. 292. Grundlehren Der Mathematischen Wissenschaften. Berlin, Heidelberg: Springer Berlin Heidelberg, 1990. ISBN: 978-3-642-08082-1 978-3-662-02661-8. DOI: 10.1007/978-3-662-02661-8. URL: http://link.springer.com/10.1007/978-3-662-02661-8 (visited on 11/09/2025).

[Rie] Emily Riehl. "Category Theory in Context". In: ().

[Sch11] Pierre Schapira. *Algebra & Topology*. Sept. 1, 2011.

[Sch] Pierre Schapira. "An Introduction to Categories and Sheaves". In: ().

[Sch19] Pierre Schapira. *Categories and Homological Algebra*. Sept. 1, 2019. URL: https://perso.imj-prg.fr/pierre-schapira/wp-content/uploads/schapira-pub/lectnotes/HomAl.pdf (visited on 11/09/2025).

[Tu22] Loring W. Tu. *Introduction to Sheaf Cohomology*. June 16, 2022. DOI: 10.48550/arXiv.2206.07512. arXiv: 2206.07512 [math]. URL: http://arxiv.org/abs/2206.07512. Pre-published.

[Vak25] Ravi Vakil. *RISING SEA: Foundations of Algebraic Geometry*. S.l.: PRINCETON UNIVERSITY PRES, 2025. ISBN: 978-0-691-26866-8. URL: https://math.stanford.edu/~vakil/216blog/FOAGsep0824public.pdf.

[Wed16] Torsten Wedhorn. *Manifolds, Sheaves, and Cohomology*. Wiesbaden: Springer Fachmedien Wiesbaden, 2016. ISBN: 978-3-658-10632-4 978-3-658-10633-1. DOI: 10.1007/978-3-658-10633-1. URL: https://sci-hub.ru/10.1007/978-3-658-10633-1 (visited on 06/24/2025).

[ZT14] Fouad El Zein and Loring W. Tu. *From Sheaf Cohomology to the Algebraic de Rham Theorem*. Jan. 14, 2014. DOI: 10.48550/arXiv.1302.5834. arXiv: 1302.5834 [math]. URL: http://arxiv.org/abs/1302.5834. Pre-published.